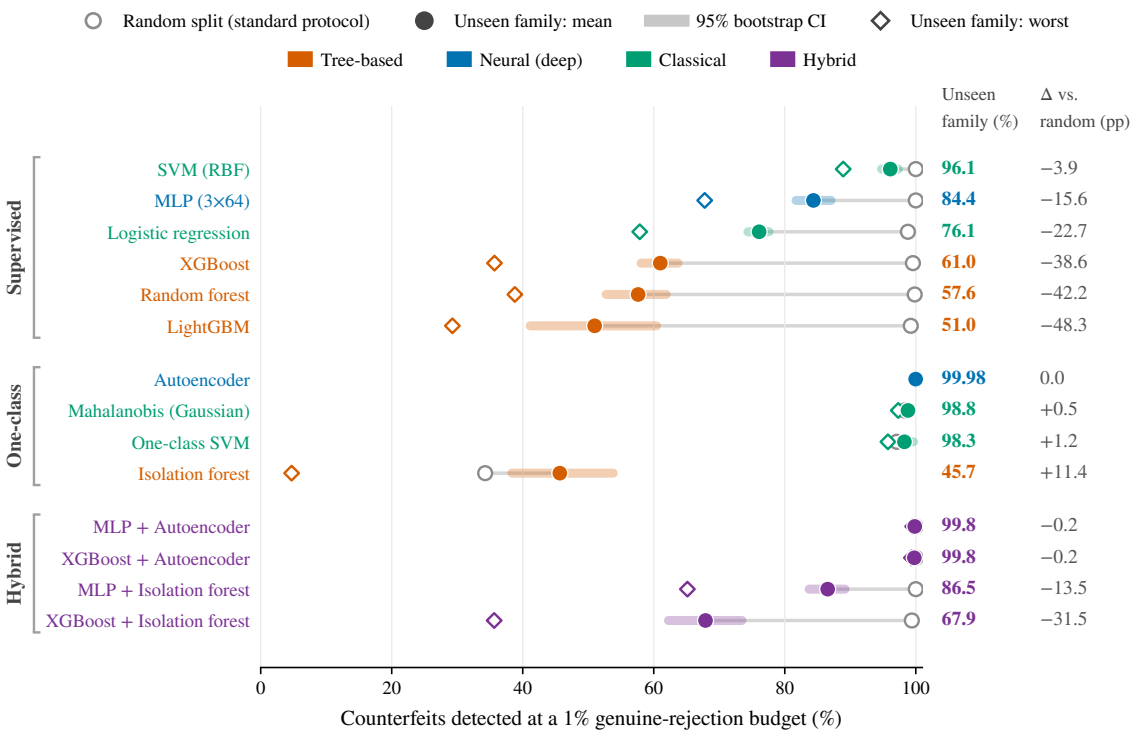
Random split (standard protocol)
Unseen family: mean
95% bootstrap CI
Unseen family: worst
Tree-based
Neural (deep)
Classical
Hybrid
Unseen family (%)
Δ vs. random (pp)
Supervised
SVM (RBF)
MLP (3×64)
Logistic regression
XGBoost
Random forest
LightGBM
96.1
84.4
76.1
61.0
57.6
51.0
−3.9
−15.6
−22.7
−38.6
−42.2
−48.3
One-class
Autoencoder
Mahalanobis (Gaussian)
One-class SVM
Isolation forest
99.98
98.8
98.3
45.7
0.0
+0.5
+1.2
+11.4
Hybrid
MLP + Autoencoder
XGBoost + Autoencoder
MLP + Isolation forest
XGBoost + Isolation forest
99.8
99.8
86.5
67.9
−0.2
−0.2
−13.5
−31.5
0
20
40
60
80
100
Counterfeits detected at a 1% genuine-rejection budget (%)